

SEX DIFFERENCES IN HUMAN SKELETAL MUSCLE ACROSS THE ADULT LIFESPAN: FROM PHENOTYPE TO TRANSCRIPTOME

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INTRODUCTION

Skeletal muscle mass and function decline with age, contributing to reduced strength, mobility and quality of life. Males and females differ not only in baseline muscle phenotype but potentially in how muscle changes across the lifespan. Understanding these sex-specific trajectories, and their molecular basis, is important for developing targeted strategies against age-related muscle loss. However, most prior work has compared discrete young/old groups rather than modelling change continuously across adulthood. How sex differences in muscle phenotype and gene expression evolve across the full adult lifespan therefore remains poorly understood.

METHODS

We studied 177 healthy adults (96 females, 81 males) aged 18–80 years to determine sex-specific changes in muscle phenotype and gene expression across the adult lifespan. Body composition was assessed via dual-energy X-ray absorptiometry, muscle morphology via peripheral quantitative computed tomography, and muscle strength via leg press one-repetition maximum. A muscle biopsy was taken from the vastus lateralis to assess muscle fibre type (immunohistochemistry) and gene expression (RNA sequencing) to explore sex specific, non-linear changes in gene expression with age.

RESULTS AND DISCUSSION

At baseline, males had greater appendicular lean mass, muscle cross-sectional area, strength and fibre size than females, along with a higher proportion of Type IIa and lower proportion of Type I fibres. With age, both sexes showed similar significant declines in muscle cross-sectional area and strength, and a significant rise in intramuscular fat, but appendicular lean mass declined

more steeply in males. Type I fibre size declined significantly with age in females, while Type IIa fibre size declined significantly with age in males but not females.

Transcriptomic analysis identified 1,193 genes displaying non-linear sex-specific changes in expression across age. Clustering of these non-linear trajectories identified 16 distinct patterns of sex-specific transcriptional ageing. Characterisation of these clusters revealed differences in their gene biotype, chromosomal representation and enriched biological processes, demonstrating that sex differences in skeletal muscle ageing extend beyond gross muscle phenotype to distinct molecular trajectories.

These findings show that sex differences in skeletal muscle ageing are selective rather than global: some traits age similarly in males and females, while others follow distinct sex-specific trajectories, and this pattern is mirrored, and considerably amplified, at the molecular level.

CONCLUSIONS

Skeletal muscle ageing is not a uniform process in males and females. Although males and females share several phenotypic patterns of muscle ageing, the underlying mechanisms driving age-related muscle loss appear to differ, and to vary by sex. Moreover, these trajectories are non-linear, underscoring that muscle ageing cannot be fully captured by comparing discrete young and old groups alone. By characterising how these sex-specific changes unfold across the ageing process, this study contributes novel insight into the biological mechanisms underlying sex differences in muscle loss during ageing.